

Ankush Makhijani

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Education

San Jose State University | US

December 2027

Masters of Science | **Major: Software Engineering** | **Sp: AI & Distributed Systems**

Rutgers University, College of Arts and Science | US

May 2025

Bachelor of Science | **Major: Computer Science**

Granted with the Chancellor's Merit Scholarship at Rutgers University.

Dean's List

Work Experience

AceMyTravel (Stealth Startup) | Remote

June 2025 – December 2025

Software Engineer Fellowship | Python, FastAPI, Docker, SQL, GCP NLP, Lang chain

- Designed and prototyped an AI-powered itinerary generation system product using LangChain prompt flows and lightweight agents to convert user prompts and live travel data into personalized tours.
- Took initiative to create a config interface in Python with unit and integration tests that reduced Database migrations and schema changes from hours to minutes.
- Integrated Google Cloud NLP APIs for semantic analysis and search, improving recommendation relevance by 28%.
- Working in cross-team collaboration teams to design, prototype, and deliver customer-facing features. Optimized performance with caching, async processing, and reactive design, reducing average response time by 19%.

Simplified IT Solutions | Thane, India

April 2024 –August 2024

Software Intern | C++, MySQL, REST API, Crow, Swagger

- My key role was in the development and maintenance of middle-tier services of ICICI pru bank using C++ and Crow, REST APIs to support a scalable web application architecture.
- Practiced Test-Driven Development (TDD) with GoogleTest, improving release stability and reducing regressions.
- Validated data contracts between APIs and frontend components, detecting several critical bugs early in the development cycle, preventing production issues
- Interfaces with clients on general and specific issues within area of expertise.

Skills

- AI & Machine Learning: LangGraph, TensorFlow, Pytorch, RAG, MCP, n8n, Agentic workflow, Claude SDK
- Programming Languages: Python, Java, C++, JavaScript, SQL, TypeScript, GO
- Cloud: AWS (Lambda, DynamoDB, S3, EC2, RDS, API Gateway, CloudWatch), GCP (compute Engine, cloud SQL, NLP)
- Database: MySQL, PostgreSQL, MongoDB
- Backend: Crow, Spring Boot, Node.js, Drogon, Flask, Fast API, Django, Redis
- CI /CD: GitLab, Github Actions, Jenkins, Docker, Kubernetes, Terraform, Jira
- Software Testing: GoogleTest, JUnit, Mockito, PyUnit, Selenium, Playwright

Projects

Outbound Agent (RU Hackathon Award Winner) | C++, Python, AWS, Kubernetes, MongoDB, Drogon, PostgreSQL

- Architected and built an event-driven microservices platform automating the full B2B outreach pipeline, from lead discovery to personalized email messaging, reducing manual effort and increasing team efficiency by 76% in manpower hours.
- Designed and developed customer facing Docker software loader in python with unit test and pipeline, users now load 20% faster than previous version.
- A distributed architecture on Kubernetes, enabling horizontal scaling of individual microservices independently based on load, with a core backend in C++, for high performance, low latency event processing.

RAG | Python, PostgreSQL-V, Gemini,

- Built a retrieval-augmented question-answering system for image/slide PDFs, combining VLM-based extraction, vector search, and grounded generation with page-level citations. Architected to scale across large document collections.
- Designed a unified Postgres schema (Postgres-V + JSONB) storing embeddings and structured table data in a single transaction, avoiding cross-store consistency issues common in split vector-DB/relational architectures
- Engineered three interchangeable chunking strategies (page-level, sentence-level, and context-augmented sentence chunking) to address dilution and topic-loss failure modes in dense, stat-heavy slides
- Improved retrieval precision by prepending an LLM-generated topic header to each chunk's embedding input at ingestion, improving accuracy by 42%